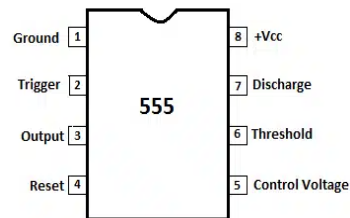
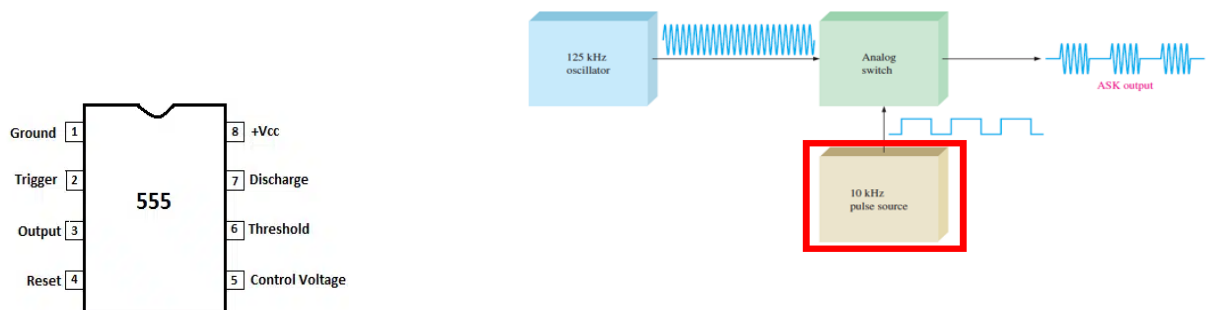


Q1. What are the output voltages of a 7809 and a 7915?

Q2. Draw the circuit of a one-shot using a 555 timer to provide one time period of 20ms. If $RA = 10k\Omega$, what value of C is needed?



Q3. The Basic block diagram of the ASK test generator in the figure below. The source must produce a 125 kHz signal that is modulated with a 10 kHz pulse signal to simulate the RFID tag. (A) **Design a pulse generator with the frequency of 10 kHz and 50% duty** cycle using 555 timer. The output of your circuit should equal to $f \cong \frac{1.44}{(RA+2RB)C}$.



(B) Modify your circuit to produce a frequency variation about 10 Hz to 10kHz to handle the source change

Q4. Using an **LM317**, design a circuit that will provide an adjustable output. Assuming transformer is specified to have a 40 V rms secondary voltage for the standard 120 V across the primary. The output can be varied from 1.2 V to 30 V depending on the resistor values.